



Modelling Financial Markets

Chapter 2: Market Efficiency — From EMH to Adaptive Markets



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AMH: Adaptive Markets Hypothesis
- ▣ BDS: Brock–Dechert–Scheinkman (test)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BIST: Borsa İstanbul (BIST 100 index) (the Istanbul Stock Exchange (BIST 100 index))
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CD: Chow–Denning (multiple variance-ratio test)
- ▣ CI: Confidence Interval
- ▣ CME: Chicago Mercantile Exchange
- ▣ CRSP: Center for Research in Security Prices
- ▣ DFA: Detrended Fluctuation Analysis
- ▣ EMH: Efficient Market Hypothesis
- ▣ ETF: Exchange-Traded Fund
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HK: Hong Kong
- ▣ IPC: Índice de Precios y Cotizaciones (the Mexican Stock Exchange index)

Acronyms used in this material (2/2)

- ▣ MDH: Martingale Difference Hypothesis
- ▣ MIT: Massachusetts Institute of Technology
- ▣ ML: Machine Learning
- ▣ MSPE: Mean Squared Prediction Error
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ RW1: Random Walk 1 (i.i.d. increments)
- ▣ RW2: Random Walk 2 (independent increments)
- ▣ RW3: Random Walk 3 (uncorrelated increments)
- ▣ SE: Standard Error
- ▣ SPA: Superior Predictive Ability (test)
- ▣ TOM: Turn Of the Month
- ▣ TSMOM: Time-Series MOMentum
- ▣ US: United States
- ▣ VAR: Vector AutoRegression
- ▣ VR: Variance Ratio
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Outline

Motivation

The Efficient Market Hypothesis

Testing Weak-Form Efficiency

Return Predictability from Valuation Ratios

Long Memory: Hurst, R/S and DFA

Adaptive Markets and Time-Varying Efficiency

Semi-Strong Efficiency: Event Studies

Anomalies

Behavioural Finance and Limits to Arbitrage

Efficiency and Machine Learning

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AI for Scientific Discovery

Conclusions

References

Can Anyone Beat the Market?

The question of this chapter

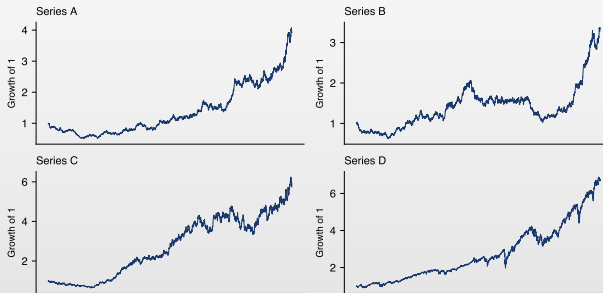
- If past prices revealed a strategy with positive expected risk-adjusted profit after costs, traders would exploit it until it disappeared
- A market in which this has already happened is called **efficient**
- The claim is testable: it restricts the statistical behaviour of returns

What we do in this chapter

- Define efficiency precisely (Fama's three forms)
- Test it: autocorrelation, runs, variance ratios, long memory
- Ask whether efficiency changes over time (adaptive markets)
- Examine anomalies, behavioural finance and machine learning

A Game: Which Series Is the Real Market?

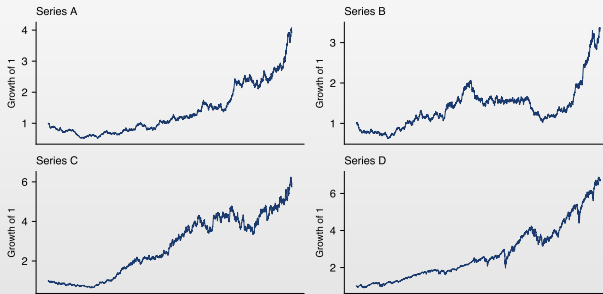
One of these is the S&P 500 (2010-2026); three are simulated random walks with the same drift and volatility



□ One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?

A Game: Which Series Is the Real Market?

One of these is the S&P 500 (2010–2026); three are simulated random walks with the same drift and volatility



- One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?
- Answer: **D. Roberts (1959)**: random walks produce “patterns” by chance; we need statistics, not eyes

Pioneers: Bachelier and Samuelson



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Paul A. Samuelson

Photo: Bernard Gotfryd (1970–1975); public domain; Wikimedia Commons

- Bachelier (1900): prices of French government bonds move like a random walk
 - ▶ “The mathematical expectation of the speculator is zero”
- Samuelson (1965): if prices are properly anticipated, their changes are unpredictable
 - ▶ Unpredictability is a sign of a well-functioning market, not of chaos
- Both results are the ancestors of the Efficient Market Hypothesis

Data Used in This Chapter

Group	Series	Period
Developed equity	S&P 500, Euro Stoxx 50, Nikkei 225, Hang Seng	2000–2026
Emerging/frontier equity	BET, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai	2000–2026
Crypto-assets	Bitcoin, Ethereum	2014/2016–2026
Exchange rate	EUR/RON, BNR reference rate	2005–2026

- Daily closing values up to 18 September 2026; returns are daily log returns $r_t = \ln P_t - \ln P_{t-1}$
- Equity indices: weekdays only; a close equal to the previous close is treated as a holiday filled with the previous price and removed
 - ▶ The rule also drops the rare genuine zero-return day (S&P 500: 3 of about 6 700 days), which moves its test statistics by less than 0.05
- Crypto trades 7 days a week: 365 observations per year
- Each series is analysed on its own calendar; no series is joined to another in this chapter

Eugene Fama and the Definition of Efficiency



Eugene F. Fama, Nobel Prize 2013

Photo: [Bengt Nyman \(2013\)](#); CC BY 2.0; [Wikimedia](#)

[Commons](#)

- [Fama \(1970\)](#): a market is **efficient** if prices “fully reflect” all available information
 - ▶ Information set $\mathcal{I}_t$: everything known at time t
- Formally: no strategy using $\mathcal{I}_t$ earns returns above the fair (risk-adjusted) return
- Nobel Prize 2013, shared with Robert Shiller (sceptic, Section 9) and Lars Peter Hansen
- Update and survey: [Fama \(1991\)](#); accessible critique: [Malkiel \(2003\)](#)

Three Forms of Efficiency

Form	Information set $\mathcal{I}_t$	Typical test
Weak	Past prices and returns	Autocorrelation, runs, variance ratios (Sections 3, 5, 6)
Semi-strong	All public information (news, reports)	Event studies (Section 7)
Strong	Public and private (insider) information	Returns of insiders and fund managers

- Each form contains the previous one: strong $\Rightarrow$ semi-strong $\Rightarrow$ weak
- This chapter focuses on the weak form; Chapter 3 tests risk-adjusted returns with factor models

Efficiency as a Fair Game: the Martingale

- **Fair game** for excess returns: $\mathbb{E}[r_{t+1} - \mu_t \mid \mathcal{I}_t] = 0$, with μ_t the required (equilibrium) return
 - ▶ With a constant required simple return m , the total-return price discounted at m , $P_t/(1+m)^t$, is a martingale [Campbell, Lo & MacKinlay \(1997\)](#); discounting at the risk-free rate gives a martingale only under the risk-neutral measure
 - ▶ P_t itself is one only if $m = 0$: a zero expected **simple** return, not a zero expected log return
- “ $\mathbb{E}[P_{t+1} \mid \mathcal{I}_t] = P_t$ ” and “ $\mathbb{E}[\ln(P_{t+1}/P_t) \mid \mathcal{I}_t] = \mu$ ” are different restrictions
 - ▶ They differ by discounting and by the Jensen term: $\mathbb{E}[\ln(1+R)] \approx \mathbb{E}[R] - \text{Var}(R)/2$
- Consequence: $\text{corr}(\varepsilon_{t+1}, g(\mathcal{I}_t)) = 0$ for any function g of the past, $\varepsilon_{t+1} = r_{t+1} - \mu_t$
 - ▶ In particular all autocorrelations $\rho_k = \text{corr}(r_t, r_{t-k})$, $k \geq 1$, are zero when $\mu_t = \mu$
- The martingale says nothing about the variance: volatility may be predictable (Chapter 1) without contradicting efficiency

Random Walk Hypotheses RW1, RW2, RW3

- Log price $p_t = \ln P_t$ with $p_t = \mu + p_{t-1} + \varepsilon_t$; the three versions differ in what they assume about ε_t [Campbell, Lo & MacKinlay \(1997\)](#)
- **RW1:** ε_t i.i.d. (the strongest; rules out volatility clustering)
 - ▶ Rejected by the stylised facts of Chapter 1
- **RW2:** ε_t independent but not identically distributed (variance may change)
- **RW3:** ε_t uncorrelated: $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$, while $\text{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) \neq 0$ is allowed
 - ▶ The relevant hypothesis for real returns: tests must be robust to heteroskedasticity

The Joint-Hypothesis Problem

- ▣ “Fair return” requires a model of expected returns (e.g. CAPM, Chapter 3)
- ▣ Every test of efficiency is a **joint** test of efficiency and of the asset-pricing model [Fama \(1991\)](#)
 - ▶ A rejection may mean an inefficient market **or** a wrong model of risk
- ▣ Example: a strategy earns 5% per year above the index; it may simply carry more risk than the index
- ▣ At daily horizons expected returns are tiny, so the choice of model matters little: weak-form tests are the cleanest
- ▣ Efficiency can never be “proved”, only not rejected by a given test on given data

The Grossman–Stiglitz Paradox

- Gathering information is costly (analysts, data, computing)
- If prices reflected **all** information, nobody would be paid for collecting it [Grossman & Stiglitz \(1980\)](#)
 - ▶ Then nobody would collect it, and prices could not reflect it
- Resolution: an **equilibrium degree of inefficiency**, just large enough to reward informed traders for their costs
- Markets with cheap information and low costs (large, liquid) should be closer to efficiency
- A testable prediction for our comparison of developed, emerging and crypto markets (Section 3)

What the EMH Does Not Say

- It does not say prices are “correct” at every moment, only that errors are not predictable
- It does not say nobody beats the market: some will, by luck; the question is whether it persists
- It does not forbid predictable volatility: RW3 allows volatility clustering
- It does not forbid predictable **expected** returns if they compensate time-varying risk
- It does not say bubbles are impossible; it says they are hard to time (Chapter 17)

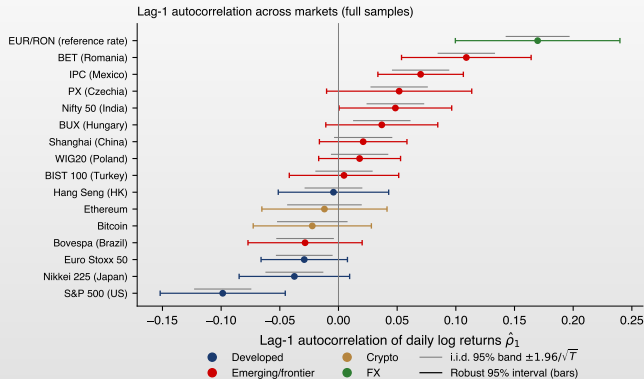
Recap: the Efficient Market Hypothesis

- Efficient market: prices fully reflect the information set $\mathcal{I}_t$ (weak, semi-strong, strong)
- Martingale / fair game: excess returns are unpredictable from $\mathcal{I}_t$
- RW3 is the version to test; RW1 is already rejected
- Every test is joint with a risk model; costly information implies some inefficiency
- Small autocorrelations can be statistically significant and economically irrelevant

Autocorrelation and Its Standard Error

- ▣ Sample autocorrelation: $\hat{\rho}_k = \sum_{t=k+1}^T e_t e_{t-k} / \sum_{t=1}^T e_t^2$, with $e_t = r_t - \bar{r}$
- ▣ Under RW1 (i.i.d.): $SE(\hat{\rho}_k) \approx 1/\sqrt{T}$; 95% band $\pm 1.96/\sqrt{T}$
 - ▶ SE: standard error, the standard deviation of an estimator
- ▣ Unpredictable mean with volatility clustering (finite fourth moments): $SE(\hat{\rho}_k) \approx \sqrt{\hat{\tau}_k}$
 - ▶ $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$ (robust variance)
 - ▶ When large moves cluster, $\hat{\tau}_k > 1/T$: the i.i.d. band is too narrow
- ▣ S&P 500: i.i.d. SE = 0.012, robust SE = 0.027, more than twice as large

Lag-1 Autocorrelation Across Markets



□ Dots: $\hat{\rho}_1$ with robust 95% intervals (bars); grey segments: the naive i.i.d. band

Lag-1 Autocorrelation: Interpretation

- ▣ Positive and significant: BET 0.109 (robust SE 0.028), IPC 0.070, EUR/RON 0.170
 - ▶ Thin trading and slow diffusion of information (BET); a managed exchange rate (EUR/RON)
- ▣ Negative and significant: S&P 500 -0.099
 - ▶ Short-term overreaction and liquidity provision; concentrated in crises (Seminar B7)
- ▣ Bitcoin -0.022 and Ethereum -0.012 : not significant with robust errors
- ▣ Several i.i.d. “rejections” disappear with robust intervals: always use robust inference
- ▣ Economic size: regressing r_t on r_{t-1} gives $R^2 = \hat{\rho}_1^2$, i.e. 0.97% (S&P 500) and 1.19% (BET) of the daily variance

Portmanteau Tests: Ljung–Box and Its Robust Version

- ▣ Ljung & Box (1978): $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k) \sim \chi_m^2$ under i.i.d. returns
- ▣ Robust version: $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k \rightarrow \chi_m^2$ under RW3 when the $\hat{\rho}_k$ are asymptotically uncorrelated (condition on the z^* slide; Chapter 1)
- ▣ $m = 10$, 5% critical value 18.31:
 - ▶ S&P 500: $Q = 96.3$ but $\tilde{Q} = 19.9$ ($p = 0.031$)
 - ▶ BET: $Q = 108.0$, $\tilde{Q} = 24.4$ ($p = 0.007$)
 - ▶ Bitcoin: $Q = 20.2$ ($p = 0.028$) but $\tilde{Q} = 11.9$ ($p = 0.294$)
- ▣ The classical test rejects too often; the robust test retains the rejections for S&P 500 and BET, but not for Bitcoin

The Runs Test

- Nonparametric: uses only the **signs** of returns (zero returns dropped), so it is immune to heavy tails [Wald & Wolfowitz \(1940\)](#)
- A **run** is a maximal sequence of equal signs: ++-+++ has 3 runs
- With n_1 positive and n_2 negative signs, $n = n_1 + n_2$, under independence:
 - ▶ $\mathbb{E}[R] = \frac{2n_1n_2}{n} + 1$, $\text{Var}[R] = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n - 1)}$
 - ▶ $z = (R - \mathbb{E}[R]) / \sqrt{\text{Var}[R]} \approx N(0, 1)$
- Too **few** runs ($z < 0$): signs persist (positive dependence of signs, not necessarily of returns); too **many** ($z > 0$): sign reversals

Runs Test on Real Markets

- ▣ S&P 500: $z = 4.34$: **more** runs than expected, consistent with $\hat{\rho}_1 < 0$
- ▣ BET: $z = -6.05$: **fewer** runs, signs persist, consistent with $\hat{\rho}_1 > 0$
- ▣ Nifty 50: $z = -5.46$; IPC: $z = -3.27$: persistence in emerging markets
- ▣ Bitcoin $z = 2.68$, Ethereum $z = 3.91$: significant although $\hat{\rho}_1$ is not
 - ▶ Signs and magnitudes carry different information: small reversals, large moves uncorrelated
- ▣ Caveat: the runs test assumes a constant probability of a positive day

The Variance Ratio: Idea

- Uncorrelated increments with constant variance: variance grows linearly with the horizon,
 $\text{Var}(r_t + \dots + r_{t-q+1}) = q \text{Var}(r_t)$
 - ▶ q : the horizon in days (e.g. $q = 5$ is one trading week)
 - ▶ With changing variances (RW2, RW3) the variance of the sum is the sum of the variances; the sample $\widehat{\text{VR}}(q)$ still tends to 1 under the conditions of [Lo & MacKinlay \(1988\)](#)
- **Variance ratio:** $\text{VR}(q) = \frac{\text{Var}(r_t + \dots + r_{t-q+1})}{q \text{Var}(r_t)}$, equal to 1 for any covariance-stationary uncorrelated series
- For a covariance-stationary series, a weighted sum of autocorrelations:
 - ▶ $\text{VR}(q) = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho_k$
- $\text{VR}(q) > 1$: momentum (positive autocorrelation); $\text{VR}(q) < 1$: mean reversion
- Worked check: S&P 500 $1 + \hat{\rho}_1 = 0.901$ vs estimated $\text{VR}(2) = 0.901$

The Lo–MacKinlay Test

- Lo & MacKinlay (1988): overlapping q -day sums, with bias corrections

▶ $\hat{\sigma}_a^2 = \frac{1}{T-1} \sum_t (r_t - \hat{\mu})^2$, $\hat{\sigma}_c^2(q) = \frac{1}{m} \sum_{t=q}^T \left(\sum_{j=0}^{q-1} r_{t-j} - q\hat{\mu} \right)^2$

▶ $m = q(T - q + 1)(1 - q/T)$; $\widehat{VR}(q) = \hat{\sigma}_c^2(q)/\hat{\sigma}_a^2$

- Under RW1 (homoskedastic):

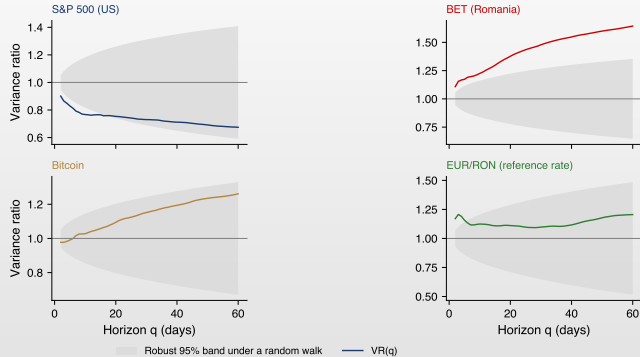
▶ $z(q) = \frac{\widehat{VR}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qT)}} \approx N(0, 1)$

- Homoskedastic: constant variance; this z over-rejects when volatility clusters

The Heteroskedasticity-Robust Statistic z^*

- Under RW3 plus the conditions of [Lo & MacKinlay \(1988\)](#) (finite moments, mixing, $\mathbb{E}[\varepsilon_t^2 \varepsilon_{t-j} \varepsilon_{t-k}] = 0$ for $j \neq k$), the $\hat{\rho}_k$ are asymptotically uncorrelated with variances $\hat{\delta}_k / T$
 - ▶ $\hat{\delta}_k = T \sum_{t=k+1}^T e_t^2 e_{t-k}^2 / \left(\sum_{t=1}^T e_t^2 \right)^2$
- Asymptotic variance of $\widehat{VR}(q)$ and the test statistic:
 - ▶ $\hat{\theta}(q) = \sum_{k=1}^{q-1} \left(\frac{2(q-k)}{q} \right)^2 \hat{\delta}_k, \quad z^*(q) = \frac{\sqrt{T} (\widehat{VR}(q) - 1)}{\sqrt{\hat{\theta}(q)}} \approx N(0, 1)$
- z^* tests RW3 under these conditions: they hold for GARCH with symmetric innovations, but can fail with a leverage effect
 - ▶ Without the last condition, $\hat{\theta}(q)$ needs the full covariance matrix of the $\hat{\rho}_k$
- S&P 500, $q = 2$: $\widehat{VR} = 0.901$, $z = -8.12$ but $z^* = -3.64$
 - ▶ Still significant, but the homoskedastic statistic is more than twice as large in absolute value

Variance-Ratio Profiles



□ $\widehat{VR}(q)$ for $q = 2, \dots, 60$ days; grey: robust 95% band around 1 (from $\hat{\theta}(q)$)

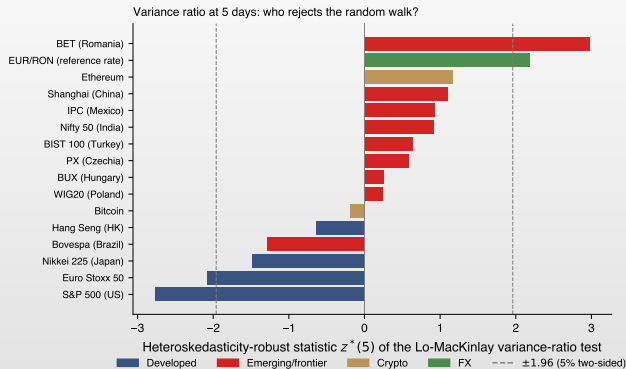
Variance-Ratio Profiles: Interpretation

- ▣ S&P 500: $VR = 0.901, 0.830, 0.753, 0.674$ at $q = 2, 5, 20, 60$: mean reversion at all horizons
- ▣ BET: 1.108, 1.173, 1.377, 1.644: persistence that accumulates with the horizon
- ▣ Bitcoin: 0.977, 0.991, 1.094, 1.261: inside the band up to about one month
- ▣ EUR/RON: 1.170, 1.157, 1.111: short-run persistence of a managed float
- ▣ The band widens with q : long horizons are informative about direction but tests there have low power

Multiple Horizons: Chow–Denning and Choi

- ▣ Testing $q = 2, 5, 10, 20$ separately inflates the probability of a false rejection
 - ▶ Four tests at 5% each: up to $1 - 0.95^4 = 18.5\%$ if independent
- ▣ [Chow & Denning \(1993\)](#): use $CD = \max_i |z^*(q_i)|$ and the studentised maximum modulus distribution
 - ▶ For $m = 4$ horizons, the 5% critical value is 2.49 instead of 1.96
- ▣ [Choi \(1999\)](#): an automatic VR test that chooses the horizon from the data (spectral estimate at frequency zero)
- ▣ Wild-bootstrap versions for small samples: [Kim \(2006\)](#); applications to Asian markets: [Kim & Shamsuddin \(2008\)](#)

Variance Ratio at One Week, Across Markets



Robust $z^*(5)$ of the Lo-MacKinlay test for all 16 series; dashed lines: ± 1.96

Variance Ratio Across Markets: Interpretation

- Reject at 5% with $z^*(5)$: BET (2.98), EUR/RON (2.19), S&P 500 (-2.77), Euro Stoxx 50 (-2.08)
- Chow–Denning (horizons 2, 5, 10, 20) rejects for 5 of 16 series
 - ▶ BET $p = 0.001$, S&P 500 $p = 0.001$, IPC $p = 0.001$, EUR/RON $p = 0.000$, Ethereum $p = 0.027$
- Bitcoin: $z^*(5) = -0.19$, Chow–Denning $p = 0.835$: no evidence against RW3 over 2014–2026
- No simple ranking “developed = efficient, emerging = inefficient”: the direction differs (reversal vs persistence)

Summary Table: Weak-Form Tests, 2000–2026

Market	T	$\hat{\rho}_1$	robust SE	$\widehat{VR}(5)$	$z^*(5)$	CD p	Runs z
S&P 500 (US)	6 714	−0.099	0.027	0.830	−2.77	0.001	4.34
Euro Stoxx 50	6 836	−0.029	0.019	0.910	−2.08	0.097	3.41
Nikkei 225 (Japan)	6 545	−0.038	0.024	0.923	−1.48	0.317	1.97
Hang Seng (HK)	6 583	−0.004	0.024	0.967	−0.64	0.872	−0.17
BET (Romania)	6 670	0.109	0.028	1.173	2.98	0.001	−6.05
WIG20 (Poland)	6 685	0.018	0.018	1.009	0.24	0.777	2.25
BUX (Hungary)	6 666	0.037	0.024	1.013	0.26	0.427	−0.42
PX (Czechia)	6 682	0.052	0.032	1.041	0.59	0.350	−2.53
BIST 100 (Turkey)	6 692	0.005	0.024	1.033	0.64	0.949	−0.63
Bovespa (Brazil)	6 622	−0.028	0.025	0.928	−1.29	0.521	0.88
IPC (Mexico)	6 689	0.070	0.019	1.039	0.93	0.001	−3.27
Nifty 50 (India)	6 606	0.048	0.024	1.048	0.91	0.178	−5.46
Shanghai (China)	6 474	0.021	0.019	1.046	1.11	0.409	1.02
Bitcoin	4 384	−0.022	0.026	0.991	−0.19	0.835	2.68
Ethereum	3 913	−0.012	0.027	1.067	1.17	0.027	3.91
EUR/RON (reference rate)	5 352	0.170	0.036	1.157	2.19	0.000	−1.66

□ T : number of daily returns; CD: Chow–Denning over $q = 2, 5, 10, 20$; EUR/RON from July 2005, crypto on 365 days per year

Beyond Linear Predictability: the Martingale Difference Hypothesis

- **MDH:** $\mathbb{E}[r_t | \mathcal{I}_{t-1}] = \mu$; RW3 (all $\rho_k = 0$) is weaker: $r_t = \varepsilon_t + \theta(\varepsilon_{t-1}^2 - \sigma^2)$ has $\rho_k = 0$ (ε_t i.i.d. symmetric) but a predictable mean
 - ▶ Generalised spectral test of the MDH, with wild-bootstrap p -values: [Escanciano & Velasco \(2006\)](#)
- [Escanciano & Lobato \(2009\)](#): automatic portmanteau $AQ = T \sum_{j \leq \tilde{p}} \hat{\gamma}_j^2 / \hat{\kappa}_j \rightarrow \chi_1^2$, $\hat{\gamma}_j = \frac{1}{T-j} \sum_t e_t e_{t-j}$, $\hat{\kappa}_j = \frac{1}{T-j} \sum_t e_t^2 e_{t-j}^2$ (robust to conditional heteroskedasticity); $\tilde{p} \leq 10$ maximises Q_p minus a penalty that switches from BIC ($p \ln T$) to AIC ($2p$) when some $\sqrt{T}|\hat{\rho}_j|$ is large
 - ▶ AQ: S&P 500 13.1 ($p = 0.0003$), BET 14.9 ($p = 0.0001$), Bitcoin 0.8 ($p = 0.385$); 5 of 16 series reject at 5%
- **BDS** [Brock, Dechert, Scheinkman & LeBaron \(1996\)](#) tests i.i.d., not the MDH: GARCH alone makes it reject
 - ▶ A BDS rejection is evidence against RW1, not against weak-form efficiency (RW3)

Recap: Testing Weak-Form Efficiency

- ▣ Autocorrelations, Ljung–Box and variance ratios test linear serial dependence; runs test sign ordering; the MDH also rules out nonlinear predictability of the mean
- ▣ Volatility clustering makes classical standard errors too small: use $\hat{\tau}_k$, $\tilde{Q}$ and z^*
- ▣ Several horizons need a joint test (Chow–Denning); several markets need care in reading “significance”
- ▣ S&P 500 shows reversal, BET and EUR/RON persistence; Bitcoin is not rejected on the full sample by the robust autocorrelation and VR tests
- ▣ Statistical rejection is not yet a profitable strategy: trading costs and risk decide

Predictive Regressions and the Stambaugh Bias

- Model: $r_{t+1} = \alpha + \beta x_t + u_{t+1}$, $x_{t+1} = \theta + \rho x_t + v_{t+1}$, with $\text{Cov}(u, v) = \sigma_{uv}$
 - ▶ r : excess return; x : log dividend-price ratio $dp_t = \ln D_t - \ln P_t$, persistent ($\rho \approx 1$)
- x_t is predetermined, not strictly exogenous: OLS is consistent but biased [Stambaugh \(1999\)](#)
 - ▶ Project u on v : $u = (\sigma_{uv}/\sigma_v^2)v + e$, so $\hat{\beta} - \beta = (\sigma_{uv}/\sigma_v^2)(\hat{\rho} - \rho) + \text{noise}$
 - ▶ With $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$: $\mathbb{E}[\hat{\beta} - \beta] \approx -\frac{\sigma_{uv}}{\sigma_v^2} \frac{1 + 3\rho}{T}$
- Price up $\Rightarrow dp$ down: $\sigma_{uv} < 0$, so the bias is **upward**: predictability looks stronger than it is
- Near a unit root the t -statistic is not $N(0, 1)$ either: valid inference by a Bonferroni Q -test over a confidence interval for ρ [Campbell & Yogo \(2006\)](#)

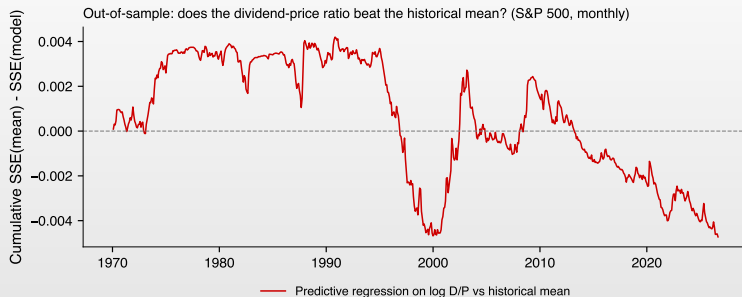
The Stambaugh Bias in Numbers: S&P 500, 1934–2026

- Monthly log excess return (dividends included, minus the 3-month T-bill rate) on lagged dp_t , $T = 1\,111$ months
 - ▶ Data: [Shiller \(2026\)](#) (S&P Composite, to 2026-06), extended with our S&P 500 closes and ETF dividends; T-bill: FRED TB3MS
- OLS: $\hat{\beta} = 0.267$ (% per month per unit of dp), Newey–West $t = 0.91$ (12 lags), $R^2 = 0.12\%$
- Predictor: $\hat{\rho} = 0.9988$; innovations: $\text{corr}(\hat{u}, \hat{v}) = -0.97$, $\hat{\phi} = \hat{\sigma}_{uv} / \hat{\sigma}_v^2 = -0.925$ (decimal units)
- Estimated bias $-100 \hat{\phi}(1 + 3\hat{\rho}) / T = 0.333$ (in %, like $\hat{\beta}$) exceeds $\hat{\beta}$ itself: bias-corrected slope -0.066
 - ▶ Corrected slope / unadjusted HAC SE = -0.22 : only illustrative, it ignores the uncertainty of the correction; formal inference: [Campbell & Yogo \(2006\)](#) or a bootstrap
- Simulation with the same ρ and $\text{corr}(u, v)$, $\beta = 0$, $T = 360$ months: the nominal 5% t -test rejects in 26.5% of samples

Out-of-Sample Predictability: R_{OS}^2 and the Clark–West Test

- Recursive forecasts $\hat{r}_t$ (expanding window) vs the historical mean $\bar{r}_t$: $R_{OS}^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t (r_t - \bar{r}_t)^2$
 - ▶ Most predictors have $R_{OS}^2 < 0$ [Welch & Goyal \(2008\)](#); the 2022 update confirms it [Goyal, Welch & Zafirov \(2024\)](#)
 - ▶ Pseudo out-of-sample, as in [Welch & Goyal \(2008\)](#): Shiller's monthly dividends are interpolated from quarterly totals, so dp_t is not strictly real-time
- Nested models: under H_0 the larger model estimates a zero β , which inflates its MSPE; Diebold–Mariano is undersized
 - ▶ [Clark & West \(2007\)](#): $f_t = (r_t - \bar{r}_t)^2 - [(r_t - \hat{r}_t)^2 - (\bar{r}_t - \hat{r}_t)^2]$; one-sided t on the mean of f_t
- [Campbell & Thompson \(2008\)](#): impose the sign of β and a non-negative forecast
 - ▶ Mean–variance timer: expected return rises by the proportion $\frac{R^2}{1 - R^2} \cdot \frac{1 + S^2}{S^2}$, S the monthly Sharpe ratio

Out-of-Sample Evidence: the Dividend–Price Ratio



- Cumulative squared-error gain over the historical mean, forecasts from 1970; rising line: the predictor helps

Out-of-Sample Evidence: Interpretation

- ▣ 680 monthly forecasts from 1970: $R_{OS}^2 = -0.51\%$; the Campbell–Thompson restrictions bind in 5.7% of months and give -0.51%
- ▣ Clark–West $t = 0.42$ (one-sided $p = 0.34$): no out-of-sample predictability from dp
- ▣ The gain peaks in 1990 and is lost in the 1990s boom, when dp fell to record lows; after 2010 it erodes again
- ▣ Economic scale: with monthly $S = 0.160$, an R^2 of only 0.5% would raise a timer's expected return by 20%
- ▣ Even a genuine slope would not reject the EMH: time-varying expected returns can be compensation for risk (joint hypothesis)

Recap: Predictability from Valuation Ratios

- ▣ Persistent predictors with innovations correlated with returns bias $\hat{\beta}$ upward and distort the t -test
- ▣ S&P 500 since 1934: the in-sample slope is insignificant and smaller than its own Stambaugh bias
- ▣ Out of sample, compare with the historical mean using R^2_{OS} and Clark–West, not Diebold–Mariano
- ▣ Small R^2 can still be economically large: judge predictability by utility, not by R^2 alone

Harold Hurst and the Nile



Harold Edwin Hurst (1880–1978)

Photo: Elliott & Fry (1953); public domain; Wikimedia

[Commons](#)

- Hurst (1951) designed Nile reservoirs: wet years followed wet years more often than chance allows, measured with the rescaled range



Nilometer on Elephantine Island, Aswan

Photo: Olaf Tausch (2011); CC BY 3.0; Wikimedia Commons

The Rescaled Range (R/S) Statistic

- For a window of n returns with mean $\bar{r}_n$ and standard deviation s_n :
 - ▶ Cumulative deviations $Y_k = \sum_{i=1}^k (r_i - \bar{r}_n)$, $k = 1, \dots, n$
 - ▶ Range $R_n = \max_k Y_k - \min_k Y_k$; rescaled range R_n/s_n
- Scaling law: $\mathbb{E}[R_n/s_n] \approx c n^H$, where H is the **Hurst exponent**
 - ▶ Estimate: slope of $\ln(R/S)$ on $\ln n$, averaging R/S over non-overlapping windows
- $H = 0.5$: no long memory (short-run dependence allowed); $H > 0.5$: persistence (long memory); $H < 0.5$: anti-persistence
 - ▶ **Long memory**: autocorrelations decay so slowly that their sum diverges, $\rho_k \sim k^{2H-2}$

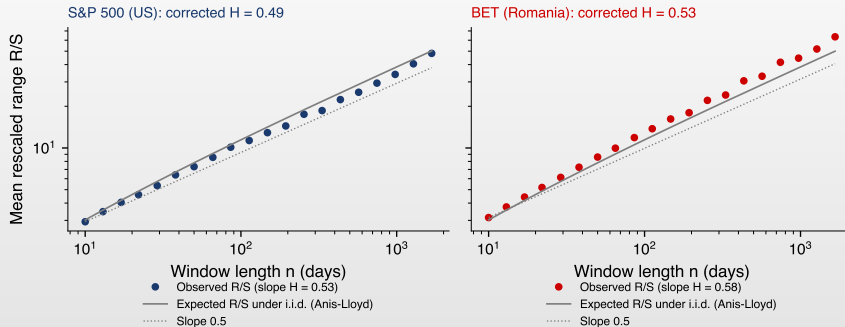
Hurst Exponent and Variance Ratio

- ▣ For self-similar increments (fractional Gaussian noise): $\text{Var}(r_t + \dots + r_{t-q+1}) = q^{2H} \text{Var}(r_t)$
- ▣ Hence $\text{VR}(q) = q^{2H-1}$ and $H = 0.5 + \ln \text{VR}(q) / (2 \ln q)$
- ▣ Worked example, Bitcoin: $\widehat{\text{VR}}(20) = 1.094$
 - ▶ $\hat{H} = 0.5 + 0.0900/5.991 = 0.515$, close to 0.5
- ▣ Same relation for BET ($\widehat{\text{VR}}(20) = 1.377$) is solved in Seminar A5
- ▣ The relation holds only for exact self-similarity: short-memory processes give $H(q)$ that depends on q

The Small-Sample Bias of R/S

- ▣ Even for i.i.d. returns, the slope of $\ln(R/S)$ on $\ln n$ exceeds 0.5 for small n
- ▣ Anis & Lloyd (1976): exact $\mathbb{E}[R_n/s_n]$ under i.i.d. Gaussian returns
 - ▶ $\mathbb{E}[R_n/s_n] = \frac{\Gamma((n-1)/2)}{\sqrt{\pi} \Gamma(n/2)} \sum_{i=1}^{n-1} \sqrt{\frac{n-i}{i}}$ (Γ : the gamma function)
- ▣ Corrected exponent: $H_{AL} = 0.5 + \text{slope of } \ln(R/S) - \ln \mathbb{E}[R_n/s_n] \text{ on } \ln n$
- ▣ Uncorrected estimates overstate memory by about 0.04–0.05 with daily data (next chart)

R/S Analysis: S&P 500 and BET



□ Dots: average R/S for window lengths n ; grey line: the Anis-Lloyd expectation under i.i.d. returns

R/S Analysis: Interpretation

- ▣ S&P 500: raw $\hat{H} = 0.530$, corrected $\hat{H}_{AL} = 0.486$: no long memory, slight anti-persistence
- ▣ BET: raw $\hat{H} = 0.578$, corrected $\hat{H}_{AL} = 0.533$: persistence remains after the correction
- ▣ Without the correction, the S&P 500 would look “persistent” ($0.530 > 0.5$): a pure small-sample artefact
- ▣ Open question: is BET's $H > 0.5$ **long** memory or just the short-run autocorrelation $\hat{\rho}_1 = 0.109$?
 - ▶ Classical R/S cannot tell them apart: next slide

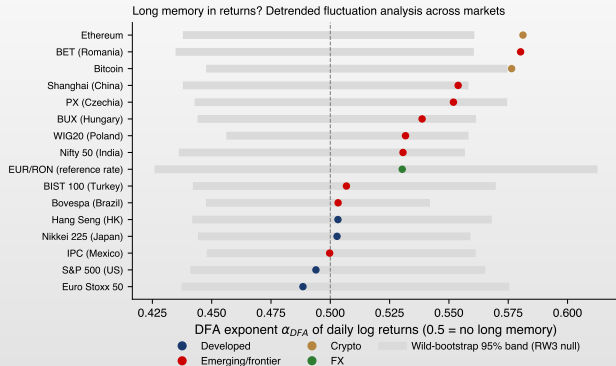
Lo's Modified R/S Test

- Lo (1991): short-run autocorrelation inflates R/S and mimics long memory
- Replace s_n by a long-run (HAC) standard deviation with ℓ lags:
 - ▶ $\hat{\sigma}_\ell^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^{\ell} \left(1 - \frac{j}{\ell+1}\right) \hat{\gamma}_j$, $\hat{\gamma}_j$: autocovariance at lag j
 - ▶ $V = \frac{\max_k Y_k - \min_k Y_k}{\hat{\sigma}_\ell \sqrt{T}}$; bandwidth ℓ chosen as in Andrews (1991)
- Under short memory, V lies in $[0.809, 1.862]$ with probability 95%
- Full samples: S&P 500 $V = 1.559$, Bitcoin 1.547, PX 1.750, EUR/RON 1.429: not rejected
 - ▶ BET $V = 1.961$ ($\ell = 7$) and Ethereum $V = 1.980$: just above 1.862, weak evidence of long memory

Detrended Fluctuation Analysis (DFA)

- Peng et al. (1994): robust to slow trends and non-stationary means
- Algorithm:
 - ▶ Profile $y_k = \sum_{i=1}^k (r_i - \bar{r})$; split into boxes of length n
 - ▶ In each box fit a linear trend and keep the residuals
 - ▶ $F(n)$ = root mean square of all residuals; repeat for many n
- $F(n) \propto n^{\alpha_{DFA}}$: the DFA exponent α_{DFA} plays the role of H ($\alpha_{DFA} = 0.5$: no long memory)
- Null band by **wild bootstrap**: $e_t \eta_t$, $\eta_t = \pm 1$ with probability 1/2 (199 replications)
 - ▶ Keeps the volatility path $|e_t|$ (allowed under RW3) and destroys serial correlation; shuffling would destroy both

DFA Exponent Across Markets



□ Dots: α_{DFA} of daily log returns; grey bars: wild-bootstrap 95% band under RW3 (volatility kept, no autocorrelation)

DFA Across Markets: Interpretation

- ▣ Inside the band: S&P 500 0.494, Euro Stoxx 50 0.488, Nikkei 0.503, PX 0.552, Shanghai 0.554
- ▣ Above the band (3 of 16): Ethereum 0.581, BET 0.580 (band up to 0.561), Bitcoin 0.576 (up to 0.575)
 - ▶ A shuffled band (upper limit about 0.55) flags 5 series: PX and Shanghai were “long memory” only through volatility clustering
- ▣ Cross-check with Lo's V : Bitcoin ($V = 1.547$) is not rejected, BET and Ethereum marginally are
- ▣ Lesson: long memory in returns is weak and fragile; long memory in volatility (Chapter 1) is strong

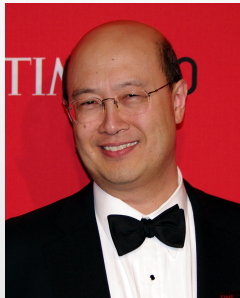
Long Memory with Inference: the Local Whittle Estimator

- $d = H - 0.5$; spectral density $f(\lambda) \sim G\lambda^{-2d}$ as $\lambda \rightarrow 0$; fit only the first m Fourier frequencies $\lambda_j = 2\pi j/T$
 - ▶ $\hat{d} = \arg \min_d \left[\ln \left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^m \ln \lambda_j \right]$, I : the periodogram
 - ▶ Robinson (1995): for $|d| < 1/2$, $m \rightarrow \infty$, $m/T \rightarrow 0$, a smooth f near 0 and a linear process with conditionally homoskedastic martingale-difference innovations, $\sqrt{m}(\hat{d} - d) \rightarrow N(0, 1/4)$ without Gaussianity: a confidence interval instead of a band
- $m = T^{0.65}$, 95% CI $\hat{d} \pm 1.96/(2\sqrt{m})$:
 - ▶ S&P 500 -0.013 $[-0.069, 0.043]$; Bitcoin 0.063 $[-0.002, 0.127]$
 - ▶ BET 0.118 $[0.062, 0.174]$; Ethereum 0.112 $[0.045, 0.179]$
- Bandwidth trade-off: a large m reaches frequencies where short-run autocorrelation (BET $\hat{\rho}_1 = 0.109$) biases $\hat{d}$ upward; with $m = T^{0.5}$ BET still gives 0.135 (SE 0.056): a sensitivity check supporting, not proving, that its $d > 0$ is not only short memory (GARCH returns also strain the conditions above)

Recap: Long Memory

- ▣ The Hurst exponent measures how the range of cumulative returns scales with the horizon: $H = 0.5$ means no long-range dependence (short memory allowed)
- ▣ The classical R/S Hurst slope is biased upward over small windows: always use the Anis–Lloyd correction
- ▣ Short memory mimics long memory: Lo's V separates them
- ▣ DFA needs a null that keeps volatility clustering (wild bootstrap); the local Whittle $\hat{d}$ comes with a confidence interval
- ▣ Developed indices show no long memory in returns

Andrew Lo and the Adaptive Market Hypothesis



Andrew W. Lo, MIT

Photo: David Shankbone (2012); CC BY 3.0;

Wikimedia Commons

- Lo (2004); book-length version Lo (2017)
- Markets are an ecology of competing species (traders, funds, algorithms)
 - ▶ Species adapt by trial and error, not by perfect optimisation
- Efficiency depends on the number of competitors, the profit opportunities and the environment
- Prediction: efficiency **varies over time**, and anomalies appear, weaken and may reappear

EMH versus AMH

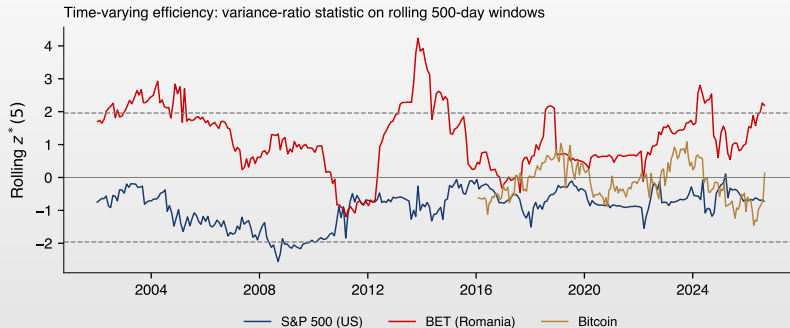
	EMH	AMH
Investors	Rational, optimising	Bounded rationality, heuristics
Efficiency	A state: yes or no	A degree that changes with conditions
Risk–return	Equilibrium risk premia, possibly time-varying	Changes with the population of investors
Anomalies	Mispriced risk or data mining	Transient profit opportunities
Empirical tool	Tests of an equilibrium restriction	Tests of time and state variation (rolling windows)

- Evidence of time variation: survey [Lim & Brooks \(2011\)](#); long-run indices [Urquhart & McGroarty \(2016\)](#)

Measuring Time-Varying Efficiency

- Compute a statistic on a **rolling window**: 500 trading days (about two years), moved forward by 21 days (one month)
 - ▶ Statistics: $z^*(5)$ of the variance-ratio test and the DFA exponent
- Null band for DFA: wild bootstrap **inside each window** (199 sign flips), so the band follows the window's volatility
 - ▶ An i.i.d. Student- t_4 band $[0.378, 0.614]$ ignores volatility clustering, which RW3 allows and which raises α_{DFA}
- Caveats
 - ▶ Overlapping windows are strongly correlated: runs of rejections are not independent evidence
 - ▶ Even under the null, about 5% of windows reject: compare the share of rejections with 5%

Rolling Variance Ratio: S&P 500, BET, Bitcoin

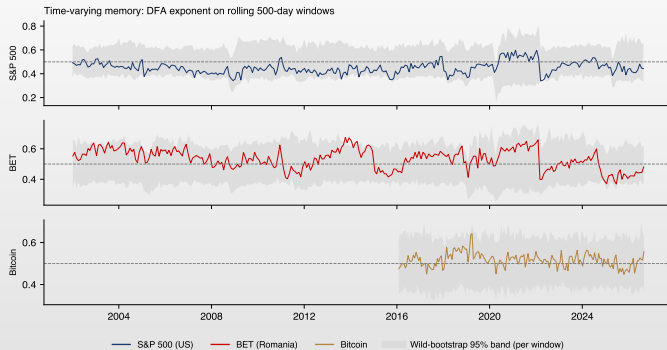


□ $z^*(5)$ on rolling 500-day windows; dashed lines ± 1.96

Rolling Variance Ratio: Interpretation

- ▣ Share of windows with $|z^*(5)| > 1.96$: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- ▣ S&P 500: close to the 5% expected by chance; the full-sample rejection is concentrated in a few turbulent windows
- ▣ BET: rejections cluster in episodes, then fade: the pattern the AMH predicts
 - ▶ Compare the pre- and post-2020 periods in Seminar C1 (FTSE upgrade)
- ▣ Bitcoin, 2014–2026: no 500-day window rejects at 5%

Rolling DFA Exponent: S&P 500, BET, Bitcoin



□ α_{DFA} on rolling 500-day windows; grey: wild-bootstrap 95% band computed in each window

Rolling DFA: Interpretation

- Windows outside their own wild-bootstrap band: S&P 500 2.0%, BET 2.7%, Bitcoin 0.0%: all below the 5% expected by chance
- With the i.i.d. Student- t_4 band: 6.4%, 12.9%, 1.1%: the apparent BET “episodes” were volatility clustering
 - ▶ The wrong null over-rejects exactly where volatility is high: a test of RW1, not of RW3
- The band is wide (average [0.36, 0.64] for BET): 500 observations cannot pin down a scaling exponent
- Time variation in efficiency is better measured by the rolling $z^*(5)$, whose null is already robust

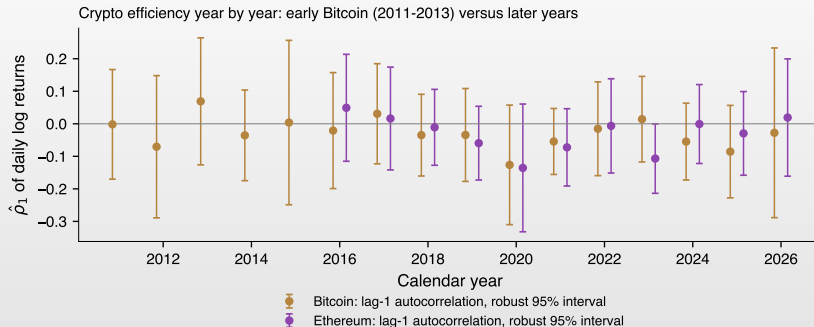
The AMH, Tested

- Kim, Shamsuddin & Lim (2011): century-long US data; predictability varies with market conditions (crises, bubbles), not along a trend
 - ▶ This turns the AMH narrative into a regression with testable coefficients
- Our version: regress the rolling $z^*(5)$ on the window's annualised volatility (%) and a crisis dummy (window overlaps Sep 2008–Jun 2009 or Feb–Jun 2020)
 - ▶ Newey–West errors with 24 lags: consecutive windows share 95% of their days
- S&P 500: crisis -0.67 ($t = -2.37$), volatility 0.006 ($t = 0.27$), $R^2 = 0.20$
- BET: crisis -0.82 ($t = -2.77$), volatility -0.008 ($t = -0.36$), $R^2 = 0.15$
- Crisis windows are associated with lower $z^*(5)$ (towards reversal) in both markets, beyond volatility: a conditional association, not a causal effect
 - ▶ z^* mixes $\widehat{\sqrt{R}} - 1$ and its standard error; state dependence is consistent with the AMH but does not identify it uniquely

Is Bitcoin Efficient? The Literature

- [Urquhart \(2016\)](#): Bitcoin inefficient over 2010–2016; some tests indicate efficiency in the later subperiod
 - ▶ His battery includes BDS, which rejects under volatility clustering alone: not a test of RW3
- [Nadarajah & Chu \(2017\)](#): with a power transformation of returns, Bitcoin is weak-form efficient
- [Kristoufek \(2018\)](#): Bitcoin markets mostly inefficient in 2010–2017, with efficiency changing over time
- [Tran & Leirvik \(2020\)](#): time-varying efficiency across many crypto-assets, increasing over time
- [Pele et al. \(2023\)](#): crypto-assets have become a distinct asset class; their returns share the stylised facts of other markets

Crypto Efficiency Year by Year



- Lag-1 autocorrelation of daily returns for each calendar year, with robust 95% intervals; Bitcoin from 2011

Crypto Year by Year: Interpretation

- ▣ Only 1 of the 27 yearly $\hat{\rho}_1$ has $|t| > 1.96$ with robust errors: Ethereum 2023, $\hat{\rho}_1 = -0.107$ (SE 0.054)
 - ▶ Chance alone would produce about $0.05 \times 27 = 1.35$ such years
- ▣ Bitcoin 2011–2013: $\hat{\rho}_1 = -0.002, -0.071, 0.069$; the largest $|z^*(5)|$ in any year is 1.71
- ▣ This differs from [Urquhart \(2016\)](#), who rejects efficiency in early years
 - ▶ Different tests (Ljung–Box, runs, Bartels, automatic VR, BDS, R/S), a different price source and robust vs classical errors
 - ▶ A single year of 365 returns gives a robust SE of about 0.06–0.13: small effects are invisible
- ▣ The evidence against crypto efficiency depends strongly on the test and on the inference

Romania: Frontier, then Emerging

- ▣ FTSE Russell (2019): Romania reclassified from Frontier to Secondary Emerging, effective 21 September 2020
- ▣ Expected channels: more foreign investors, more analysts, more liquidity $\Rightarrow$ faster price discovery
- ▣ Earlier BVB evidence: crashes and log-periodic bubbles [Pele & Mazurencu-Marinescu \(2012\)](#); herding in an illiquid market [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)
- ▣ Did efficiency improve? Seminar C1 answers with a before/after test and a block bootstrap
 - ▶ Block bootstrap: resamples blocks of consecutive days to keep the dependence [Künsch \(1989\)](#)

Recap: Adaptive Markets

- ▣ The AMH replaces “efficient or not” with “how efficient, when and why”
- ▣ Rolling tests reveal episodes: BET rejects in 22.4% of windows, the S&P 500 near the 5% chance level
- ▣ Rolling statistics need a null that keeps volatility clustering (wild bootstrap) and must be read as correlated evidence
- ▣ Crisis windows are associated with lower $z^*(5)$ (towards reversal): state dependence consistent with the AMH, not a proof of it
- ▣ Crypto efficiency is test-dependent: 1 of 27 robust yearly tests rejects (Ethereum 2023), about what chance produces
- ▣ Market-structure events (FTSE upgrade) are natural experiments for efficiency

The Event-Study Method

- ▣ Fama, Fisher, Jensen & Roll (1969): how fast do prices absorb public news (stock splits)?
- ▣ Market model on an estimation window before the event: $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$
 - ▶ $r_{m,t}$: market return; b_i : sensitivity of asset i to the market
- ▣ Abnormal return $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$; cumulative $CAR_i[\tau_1, \tau_2] = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}$
- ▣ First-pass test: $t = CAR_i / (\hat{\sigma}_\varepsilon \sqrt{L})$, with L the length of the event window MacKinlay (1997); refinements two slides ahead
- ▣ Efficiency predicts a jump on day 0 and no drift or reversal afterwards

Worked Example: the Romanian Bank Tax of December 2018

- ▣ Seminar 0, Exercise 7: tax announced on the evening of 18 Dec 2018; day 0 = 19 Dec; Banca Transilvania vs BET
- ▣ Market model on days $[-250, -11]$: $\hat{b} = 1.18$, $\hat{\sigma}_\varepsilon = 0.83\%$
- ▣ Day 0: $AR_0 = -8.3\%$ ($t = -10.0$); $CAR[0, +5] = -1.1\%$ ($t = -0.54$)
- ▣ Post-event window: $CAR[+1, +5] = +7.2\%$, $t = 3.82$ (full prediction-error covariance, next slide)
- ▣ Most of the day-0 fall reverses significantly within the week: consistent with an **overreaction**, against semi-strong efficiency
 - ▶ Or new information (the tax was later softened); a single event cannot separate the two: the joint-hypothesis problem again

Event Studies: Inference Done Properly

- Prediction-error variance: $\text{Var}(AR_\tau) = \sigma_\varepsilon^2 \left[1 + \frac{1}{T_0} + \frac{(r_{m,\tau} - \hat{\mu}_m)^2}{(T_0 - 1)\hat{\sigma}_m^2} \right]$, T_0 = estimation days [MacKinlay \(1997\)](#)
 - ▶ Bank tax, BET $r_{m,0} = -11.9\%$: the bracket is 2.12, so TLV's t shrinks in absolute value from -10.0 to -6.85 ; BRD: $AR_0 = -5.6\%$, $t = -4.03$
- Several firms: standardise each AR by its own prediction error and use the cross-sectional variance, robust to event-induced variance [Boehmer, Masumeci & Poulsen \(1991\)](#)
- Same event day for all firms: abnormal returns are cross-correlated and cross-sectional tests over-reject [Kolari & Pynnönen \(2010\)](#)
 - ▶ Correction factor $\sqrt{(1 - \bar{r}) / (1 + (N - 1)\bar{r})}$, $\bar{r}$ = average residual correlation
- TLV and BRD: $\bar{r} = 0.15$; mean AR_0 with an independence $t = -7.55$, portfolio $t = -7.03$; with ten such firms the variance of the mean AR is $1 + 9\bar{r} = 2.38$ times its independence value, and a cross-sectional test, whose variance estimate is biased down by $1 - \bar{r}$, understates it 2.81 times

How Fast Is “Fast”?

- Daily data can only show that news is absorbed within a day
- At high frequency, futures prices react to scheduled macro news within microseconds; [Tak, Schlamp, Osterrieder & Pele \(2026\)](#) measure this race at Eurex and CME
- Crypto events: [Pele, Lin, Wang & Gaman \(2025\)](#) study market responses to Ethereum development milestones
- Speed of adjustment is itself a measure of efficiency, and it depends on technology (Chapter 10, market microstructure)

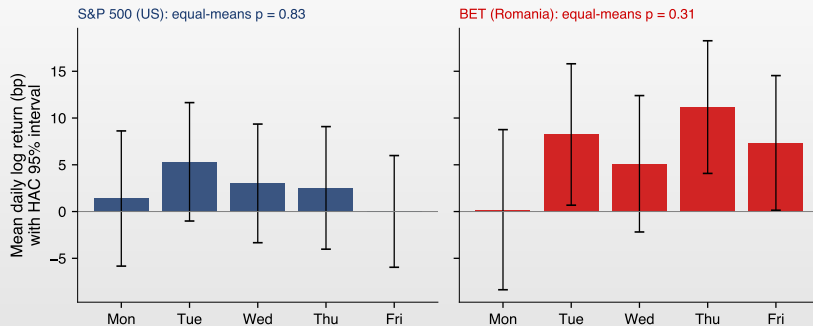
Recap: Event Studies

- Semi-strong efficiency predicts a jump on day 0 and no drift or reversal afterwards
- On a large market move, the estimation error of the market model doubles the variance of AR_0
- Clustered event days need tests that allow for cross-correlation (portfolio, Kolari–Pynnönen)
- The bank-tax fall survives every correction; the significant reversal over days 1–5 is consistent with overreaction, but one event cannot rule out later news

Anomalies and Calendar Effects

- An **anomaly** is a return pattern the asset-pricing model cannot explain: mispriced risk, behavioural bias or data mining
 - ▶ Families: calendar, time-series (index momentum), cross-sectional (size, value, momentum: **Chapter 3**)
 - ▶ Returns 26% lower out of sample and 58% lower after publication [McLean & Pontiff \(2016\)](#)
- Calendar effects: Monday [French \(1980\)](#); January [Rozeff & Kinney \(1976\)](#); turn of the month (TOM: last trading day and first three of the next month) [Ariel \(1987\)](#); [Lakonishok & Smidt \(1988\)](#)
 - ▶ Day of week: $r_t = \sum_{d=1}^5 \beta_d D_{d,t} + u_t$, Wald test of $\beta_1 = \dots = \beta_5$; TOM and January: one dummy each
 - ▶ Newey–West HAC errors [Newey & West \(1987\)](#), $L = 5$ days or 3 months: residuals are heteroskedastic and autocorrelated

Day-of-the-Week Effect: S&P 500 and BET



□ Mean daily log return by weekday (bp), 2000–2026, with HAC 95% intervals

Day of the Week: Interpretation

- ▣ S&P 500: means between 0.0 bp (Friday) and 5.3 bp (Tuesday); equal-means $p = 0.83$
- ▣ No Monday effect since 2000: the anomaly of French (1980) has disappeared
- ▣ BET: Thursday 11.2 bp ($t = 3.09$), Monday 0.2 bp; yet the joint test gives $p = 0.31$
- ▣ A single large t among five days is expected by chance: test the joint hypothesis
 - ▶ Individual t tests against zero also mix the “day effect” with the positive average return

January and Turn-of-the-Month Effects

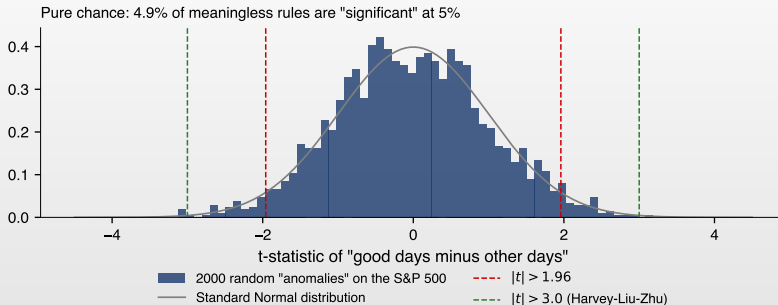
Market	January extra (%/month)	t_{HAC}	TOM extra (bp/day)	t_{HAC}	p
S&P 500 (US)	-0.66	-0.80	3.2	0.93	0.3539
BET (Romania)	3.48	1.49	17.3	3.62	0.0003
Euro Stoxx 50	-0.30	-0.29	5.1	1.21	0.2266
WIG20 (Poland)	0.53	0.40	8.2	1.73	0.0841
BUX (Hungary)	2.66	2.25	15.1	3.63	0.0003

- TOM: BET +17.3 bp per day and BUX +15.1 bp, both with $|t| > 3.5$; S&P 500 only +3.2 bp
- January: large but imprecise for BET ($t = 1.49$); BUX, with $t = 2.25$, is the only market with a January-effect p -value below 0.05
- We ran 15 tests (5 markets $\times$ 3 effects): which survive a correction for multiple testing? Next slides

Multiple Testing: Bonferroni and Holm

- With M independent tests at level α , $\text{FWER} = 1 - (1 - \alpha)^M$
 - ▶ $M = 15$, $\alpha = 5\%$: $\text{FWER} = 53.7\%$, a coin flip
- **Bonferroni**: reject only if $p_i < \alpha/M$; guarantees $\text{FWER} \leq \alpha$ for any dependence
- **Holm**: sort $p_{(1)} \leq \dots \leq p_{(M)}$; reject while $p_{(j)} < \alpha/(M - j + 1)$, stop at the first failure
 - ▶ Same guarantee, never less powerful than Bonferroni
- Our 15 calendar tests: Bonferroni threshold 0.0033; 3 raw rejections, 2 survive Holm (BET and BUX turn of the month)
- Seminar B6 carries out the full correction

How Many “Anomalies” Does Pure Chance Produce?



- 2000 meaningless rules on the S&P 500: each labels a random 20% of days as “good” and tests good minus other days

Pure Chance: Interpretation

- 4.9% of the random rules have $|t| > 1.96$, exactly the 5% the test promises; the largest $|t|$ is 3.19
- A researcher who tries many rules and reports the best one will always find a “significant anomaly”
 - ▶ Data snooping in technical trading rules: [Sullivan, Timmermann & White \(1999\)](#)
- [Harvey, Liu & Zhu \(2016\)](#): after hundreds of published factors, a new one should have $|t| > 3.0$; here 0.25% of random rules pass
- BET and BUX turn of the month ($|t| > 3.5$) survive even this stricter threshold

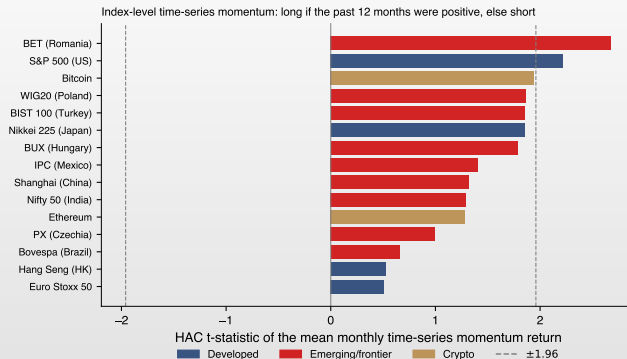
Dependent Tests: Reality Check, SPA and StepM

- Bonferroni and Holm ignore the correlation between tests (same markets, same days)
 - ▶ [White \(2000\)](#): Reality Check, bootstrap of $\max_j$ over all rules tried; [Hansen \(2005\)](#): SPA, studentised and less sensitive to poor rules
 - ▶ [Romano & Wolf \(2005\)](#): StepM, step-down control of the FWER from the **joint** bootstrap distribution of all statistics
- Our 15 calendar tests: StepM with a bootstrap of blocks of 3 calendar months, common to all markets, 999 replications
 - ▶ Critical p at the first step 0.0016 (Bonferroni: 0.0033); 2 rejections, BET and BUX TOM, as with Holm
 - ▶ Why stricter than Bonferroni? At nominal 5%, the bootstrap rejection rates of the HAC tests are 6.0% (day of week), 6.4% (TOM), 6.4% (January)
- From over two million simulated strategies: thresholds $|t| > 3.8$ (time series) and 3.4 (cross-section); ignoring them makes about 45% of rejections false [Chordia, Goyal & Saretto \(2020\)](#); lowering false discoveries raises missed ones [Harvey & Liu \(2020\)](#)

Momentum and Reversal

- **Reversal:** past long-term losers (3–5 years) outperform past winners [De Bondt & Thaler \(1985\)](#)
- **Cross-sectional momentum:** stocks that rose most over 3–12 months keep outperforming [Jegadeesh & Titman \(1993\)](#) (Chapter 3)
- **Time-series momentum (TSMOM)** of a single index [Moskowitz, Ooi & Pedersen \(2012\)](#):
 - ▶ Signal $s_t = \text{sign}\left(\sum_{j=1}^{12} r_{t-j}\right)$ from the past 12 monthly log returns
 - ▶ Strategy return $s_t R_t$, R_t the simple monthly return: long after a positive year, short after a negative one, before financing and costs
 - ▶ [Moskowitz, Ooi & Pedersen \(2012\)](#) use futures excess returns scaled to 40% ex-ante volatility; our price indices lack financing rates and dividends: a data limitation
 - ▶ Test: HAC t of the mean of $s_t R_t$ (6 lags); Sharpe ratio $\sqrt{12} \bar{x} / s_x$

Index-Level Time-Series Momentum



□ HAC t -statistic of the mean monthly TSMOM return, 2000–2026 (crypto from 2014/2016)

Time-Series Momentum: Interpretation

- ▣ Significant at 5%: BET ($t = 2.68$, 1.40% per month) and S&P 500 ($t = 2.22$)
- ▣ But the Sharpe ratio stays below buy-and-hold: BET 0.65 vs 0.75, S&P 500 0.48 vs 0.52
 - ▶ Much of the TSMOM return is the equity premium earned while long
- ▣ BET's monthly autocorrelation is 0.19: the same persistence found in daily data
- ▣ Bitcoin: $t = 1.94$; the sample contains only 307 months for equities and far fewer for crypto, limiting test power
- ▣ 15 markets tested: the multiple-testing caution applies here too

Recap: Anomalies

- Test calendar effects with regressions on dummies and HAC standard errors; test all days jointly
- The Monday effect is gone; the turn of the month survives in BET and BUX even after Holm correction
- Pure chance produces 5% “anomalies”: correct for the number of tests (Bonferroni, Holm, $|t| > 3$)
- Index momentum is statistically visible in BET and S&P 500 but does not beat buy-and-hold on a Sharpe basis
- Cross-sectional anomalies and the factor zoo: Chapter 3

Robert Shiller and Richard Thaler



Robert J. Shiller, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons



Richard H. Thaler, Nobel Prize 2017

Photo: Bengt Nyman (2017); CC BY 2.0; Wikimedia Commons

- Shiller (1981): stock prices move far more than the dividends that should justify them (excess volatility)
- De Bondt & Thaler (1985): investors overreact; extreme losers become winners over the next 3–5 years

Psychology in Prices, from the Floor to the Feed



New York Stock Exchange trading floor, 1963

Photo: Thomas J. O'Halloran (1963); public domain;
Wikimedia Commons

- **Prospect theory** [Kahneman & Tversky \(1979\)](#): losses loom larger than gains relative to a reference point
 - ▶ Disposition effect: winners sold too early, losers held too long
- **Overconfidence**: too much trading; **anchoring** and **limited attention**: slow reaction to news, a candidate cause of momentum
- **Herding** [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#): once voices on a floor, now order books, algorithms and social media (Chapter 10)
- Biases alone do not break efficiency: rational arbitrageurs could trade against them. Can they?

Limits to Arbitrage



Andrei Shleifer, Harvard

Photo: Manuel Castells (2018); CC BY-SA 2.0;

[Wikimedia Commons](#)

- **Shleifer & Vishny (1997)**: real arbitrage is risky and needs capital
- **Noise-trader risk**: mispricing can get worse before it corrects
 - ▶ Noise traders: investors trading on non-information (sentiment, rumours)
- Fund managers lose clients after losses and must close positions at the worst time
- Costs: short-selling limits, margins, trading costs, no perfect substitutes
- Hence anomalies can persist where arbitrage is hard: small, illiquid, frontier markets

Recap: Behavioural Finance

- ▣ Excess volatility and overreaction challenge the EMH; prospect theory and herding offer mechanisms
- ▣ Biases matter for prices only when arbitrage is limited by risk, capital and costs
- ▣ Where arbitrage is hardest (small, illiquid markets like BET) we indeed find the strongest persistence
- ▣ The AMH reconciles both views: rational and biased behaviour coexist and evolve

Can Machine Learning Beat Efficient Markets?

- ▣ ML models capture nonlinear patterns that autocorrelation tests miss
 - ▶ ML: machine learning, algorithms that learn a prediction rule from data (Chapter 13)
- ▣ [Gu, Kelly & Xiu \(2020\)](#): trees and neural networks give large economic gains over linear regressions in the US cross-section
- ▣ Yet their monthly out-of-sample R^2 is below 1%: consistent with an almost efficient market
- ▣ Large language models read news and filings: [Tak & Pele \(2025\)](#) forecast stock trends with LLaMA
- ▣ AMH reading: every successful model attracts competitors, and its edge decays

ML and Efficiency: Pitfalls

- ▣ **Data snooping:** many models on one dataset produce a “winner” by chance (as in our random-rules experiment)
- ▣ **Look-ahead bias:** using information not available at the time of the forecast
- ▣ **Costs:** high-turnover strategies lose their edge after trading costs
- ▣ **Non-stationarity:** a pattern found in 2000–2015 may be gone by 2026 ([McLean & Pontiff \(2016\)](#))
- ▣ Chapter 13 shows walk-forward validation and the deflated Sharpe ratio to guard against these

Big Data and Efficiency: Research Question

- [Martin & Nagel \(2022\)](#): investors price N assets whose cash flows depend on J characteristics with unknown coefficients; what does a rejected no-predictability test mean when J is comparable to N ?
 - ▶ the coefficients must be learned from past data: a high-dimensional prediction problem, as in machine learning
- Semi-strong efficiency: prices “fully reflect” public information [Fama \(1970\)](#); two readings
 - ▶ investors know the parameters of the cash-flow model (rational expectations: the null of most efficiency tests)
 - ▶ investors update their beliefs optimally (Bayes) with the data available in real time
- Why it matters: hundreds of in-sample return predictors, the “factor zoo” [Harvey, Liu & Zhu \(2016\)](#); text, satellite images and social media keep increasing J
 - ▶ does every significant predictor need a risk-based or a behavioural explanation?
- Link to this chapter: in-sample vs out-of-sample predictability [Campbell & Thompson \(2008\)](#), [Goyal, Welch & Zafirov \(2024\)](#), and the decay of anomalies after publication [McLean & Pontiff \(2016\)](#)
- Investors are risk-neutral in the model: no joint-hypothesis problem, so no predictability can be a risk premium

Big Data and Efficiency: Model and Method

- **Model** (Sec. II): cash-flow growth $\Delta y_t = Xg + e_t$, X the $N \times J$ matrix of characteristics, $e_t \sim N(0, I)$; zero interest rate
 - ▶ prior $g \sim N(0, (\theta/J)I)$: the posterior mean is a ridge regression (OLS with a penalty on the squared coefficients), i.e. coefficients shrunk towards 0 (Prop. 1)
 - ▶ realised return = underreaction caused by shrinkage + estimation noise + unpredictable shock (Prop. 2)
- **Tests**: in sample, a cross-sectional OLS of r_{t+1} on X with a joint χ^2 test; out of sample, a portfolio with weights $\frac{1}{N}Xh_{s+1}$ from coefficients estimated in other periods
 - ▶ $r_{\text{OOS},t+1} = \frac{1}{N}h'_{s+1}X'r_{t+1}$, $s \neq t$ (eq. 19)
- **Asymptotics and simulations**: $N, J \rightarrow \infty$ with $J/N \rightarrow \psi > 0$; simulations with $N = 1000$, standard-Normal entries of X and $\theta = 1$ (Sec. IV)
- **Empirical design** (Sec. VII): US stocks in CRSP, without those below the 20th NYSE size percentile or priced below \$1
 - ▶ predictors: simple and squared monthly returns at lags $t - 2$ to $t - 120$ (238 predictors), demeaned and standardised across stocks every month
 - ▶ full-sample ridge, 1971 – June 2019, penalty chosen by leave-one-year-out cross-validation; rolling 20-year OLS gives r_{OOS} from January 1956
 - ▶ figure numbers refer to the [working-paper version](#) (May 2021)

Big Data and Efficiency: Key Results

- In sample: the rejection probability of the no-predictability test tends to 1 as $N, J \rightarrow \infty$ (Prop. 4); for $\psi > 0.4$ ($\theta = 1$, $t = 1$) the probability of no rejection is of the order $e^{-0.015N}$ (Fig. II)
 - ▶ with fully rational Bayesian investors, the rational-expectations null is almost surely rejected: a rejection is not evidence of irrationality
- Simulations ($N = 1000$, four periods of learning): a test of nominal size 5% rejects with probability above 90% once $J > N/2$ (Fig. IIIb)
 - ▶ not a size distortion: the in-sample predictable component really is in the prices
- Single-predictor tests with k -FWER control: for $k = 20$ and $J/N = 0.05$, 25% rejections instead of 5%; about 80% for J/N close to 1 (Fig. IV)
 - ▶ multiple-testing corrections do not help: the problem is the null hypothesis, not the test
- Out of sample: $E[r_{\text{OOS},t+1}] = 0$, forward and backward in time (Prop. 6); positive only if investors shrink more than the data warrant (Fig. VI)
 - ▶ out-of-sample tests keep their economic meaning
- US stocks: the ridge coefficients recover momentum, long-run reversal and momentum seasonality (Fig. VII); $r_{\text{IS}} \approx 0.006\%$ per month, an in-sample R^2 of about 1%, stable over time (Fig. VIII)
 - ▶ r_{OOS} is much lower and often near 0; in the 10-year windows ending in 2014–2019 it is less than two standard errors above 0: little of the in-sample predictability survives

Big Data and Efficiency: Replicate, Limits, Extend

- **Replicate with the course data** (stock-level CRSP files are not part of them)
 - ▶ 49 industry portfolios, monthly value-weighted returns from July 1926 ([Kenneth French Data Library](#)); the same 238 predictors, monthly standardisation and rolling 20-year windows (Seminar 2, C3)
 - ▶ data limitation: 49 portfolios instead of thousands of stocks; the size and price filters do not apply to portfolios
- **Limits stated by the authors**
 - ▶ lagged returns are equilibrium outcomes, not the exogenous X of the model: the application illustrates the wedge, it does not test the model
 - ▶ a t -test on the monthly r_{OOS} is not a valid test of $E[r_{\text{OOS}}] = 0$: the expectation also averages over g
 - ▶ the backward result may rely on i.i.d. cash-flow growth; homogeneous investors and constant characteristics
- **Open extensions**
 - ▶ time series, as the authors suggest: does the dividend–price regression of this chapter show the same in-sample/out-of-sample wedge?
 - ▶ nonlinear predictors (trees, neural networks [Gu, Kelly & Xiu \(2020\)](#)) enlarge J : does the wedge grow?
 - ▶ AMH: is early out-of-sample predictability the excess shrinkage of investors with less computing power?

An Open Question: Did Spot ETFs Change Bitcoin's Efficiency?

- US spot Bitcoin ETFs trade from 11 January 2024: new arbitrageurs, new liquidity, a new venue for price discovery
- [Kia, Liu, Li, Song & Xu \(2025\)](#): with a fractionally cointegrated VAR on minute data, the ETF led the spot market in price discovery in 2021–2022
 - ▶ after the spot ETFs, the spot market became the dominant venue again
- [Tran & Leirvik \(2020\)](#): crypto efficiency varies over time and has increased: the AMH view of this chapter
- Open: does a change in market structure change weak-form efficiency, and can a before/after design identify it?
 - ▶ confounders in 2024: the April halving, the rate cycle, record inflows

The Discovery Loop, Step by Step (1/2)

□ Literature

- ▶ List peer-reviewed studies since 2020 on how ETF introductions change the efficiency or price discovery of the underlying. For each: DOI, design (event study, DiD, structural break), efficiency measure. Flag any entry you are unsure exists.
- ▶ Resolve every DOI on Crossref; the title and authors must match

□ Hypothesis

- ▶ Give three mechanisms by which spot ETFs could change Bitcoin's serial dependence (arbitrage, investor mix, trading hours). For each, the sign of the change in ρ_1 and $VR(q)$, and a variable that separates it from the others.

□ Data and code

- ▶ Read BTC-USD.CC from data/market (close); log returns; ρ_1 with a heteroskedasticity-robust standard error and the Lo-MacKinlay $VR(5)$ with z^* , in two windows of equal length around 11 January 2024.
- ▶ Replicate a known number first: the chapter's full-sample Bitcoin $\hat{\rho}_1 = -0.022$ (robust SE 0.026)

The Discovery Loop, Step by Step (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: window lengths, weekday-only closes, $VR(q)$ for $q = 2$ to 20 with Chow-Denning, placebo dates, and Ethereum as a control until its own spot ETFs (July 2024).

▣ Critique

- ▶ Act as a hostile referee: if Bitcoin was already efficient before 2024, what could this test ever detect? Compute the minimum detectable change in ρ_{01} and name a better outcome variable.

▣ Report

- ▶ Estimates, robust standard errors, the placebo distribution, every test run (with a multiple-testing correction)
- ▶ Prompts that shaped a decision go into `AI_USE.md`

- ▣ The assistant drafts the tests; the human decides which one answers the question

What the Researcher Must Check

- **i.i.d. standard errors:** $1/\sqrt{T}$ rejects too often under volatility clustering
 - ▶ test: robust SE and z^* , as in the weak-form tests of this chapter
- **Unequal windows:** different power before and after; test: equal lengths, fixed in advance
- **Event date chosen from the data**
 - ▶ test: the regulatory date only, plus a placebo distribution at dates without an event
- **Confounding:** before/after is not causal; test: a control asset (Ethereum before July 2024) in a difference-in-differences
 - ▶ causal only with parallel pre-trends, no anticipation and no spillover from Bitcoin ETFs to Ethereum; the common post-period (Jan–Jul 2024) is short; otherwise the comparison is descriptive
- **Many statistics:** ρ_1 , VR(2–20), runs; test: Chow–Denning or Holm, not the best single p -value
- **Low power:** report the minimum detectable change, ≈ 2.8 standard errors of the difference

Mini-Case: Did Bitcoin Become More Efficient?

- A typical assistant proposal: “spot ETFs brought arbitrageurs, so Bitcoin became more efficient”
- Test: two windows of 982 days around 11 January 2024; $H_0 : \rho_{1,\text{post}} = \rho_{1,\text{pre}}$
 - ▶ $z = \Delta\hat{\rho}_1 / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$, robust standard errors s
- Before (May 2021 – Jan 2024): $\hat{\rho}_1 = -0.036$ ($s = 0.042$), $\text{VR}(5) = 1.000$ ($z^* = 0.00$)
- After (Jan 2024 – Sep 2026): $\hat{\rho}_1 = -0.050$ ($s = 0.049$), $\text{VR}(5) = 0.939$ ($z^* = -0.64$); $\Delta\hat{\rho}_1 = -0.014$, $z = -0.21$
- Placebo at 11 January 2016–2021: 0 of 6 significant, largest $|z| = 1.57$
- Interpretation: neither window rejects the null; these tests do not establish whether smaller inefficiencies existed before the ETFs
 - ▶ minimum detectable change in ρ_1 : 0.18; the question needs another outcome, such as intraday price discovery

 ai_discovery_case.py

Your Turn: A Project Seed

- **Question:** did the spot ETFs change the time-varying efficiency of Bitcoin, relative to a control asset?
 - ▶ rolling VR and DFA (this chapter) for Bitcoin and Ethereum; Ethereum is untreated until July 2024
 - ▶ fix a common window ending July 2024 and discuss parallel trends, anticipation and spillovers; without them, report descriptive, not causal, differences
- **Replicate first:** the chapter's yearly Bitcoin autocorrelations and the mini-case numbers
- **Deliverables**
 - ▶ a pre-registration note (H_0 , H_1 , dates, windows, corrections) committed before the results
 - ▶ code as a Quantlet; a report with robust inference, placebo tests and power
- **Logs:** prompts in `AI_USE.md`; every caught error (an i.i.d. standard error, an invented reference) in `AI_ERRORS.md`

Conclusions

- ▣ Efficiency means unpredictable excess returns given the information set; weak-form tests are the cleanest
- ▣ Robust inference is essential: volatility clustering makes classical tests reject too often
- ▣ 2000–2026: S&P 500 shows mild reversal, BET and EUR/RON persistence, Bitcoin no robust rejection by autocorrelation and VR tests (its DFA exponent is above the band)
- ▣ Long memory in returns is weak; apparent memory is often short memory or volatility clustering
- ▣ The dividend–price slope is insignificant in sample even before the Stambaugh correction, and out of sample it does not beat the historical mean
- ▣ Efficiency varies over time (AMH); anomalies must survive multiple-testing corrections and costs
- ▣ Statistical rejection $\neq$ exploitable profit $\neq$ irrational market

Further Study on Quantinar

Advanced courses on quantinar.com

-  Efficiency of cryptocurrency markets - A GMM-based analysis
-  Generalized Method of Moments

Code, data and textbooks

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_02
- Textbooks: Campbell, Lo & MacKinlay (1997), Ch. 2; Franke, Härdle & Hafner (2019)

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